FNB eWallet to MTN MoMo

Internal implementation guide | Version 2.1

Audience: Four-member final-year Computer Science project group, University of Eswatini. **Research cut-off:** 15 October 2026.

Project team:

202405666	Musawenkosi Magongo	Project Manager	202302550	Setsembele Magagula	Designer
157538	Bandile Lokothwayo	Programmer	202302611	Sambulo Mdluli	Tester

1 What the system must do

Develop two connected simulations. In the banking application, User 1 sends money to User 2's eWallet. In a separate USSD application, two tabs represent **User 2's eWallet phone** and **User 3's MoMo phone**. USSD refers to the short text menus normally opened by dialling a code. This guide describes what the group will build; it does not include the HTML application.

Main example: User 1 sends E200 to User 2, who then sends E150 to User 3. Use test accounts, fictional phone profiles, simulated messages and simulated money. User 2 does not need an FNB bank account.

1.1 Real services and proposed features

- **Confirmed eWallet behaviour:** FNB's Eswatini information describes sending from a bank account to an eWallet without requiring the recipient to have a bank account (First National Bank of Eswatini [FNB], n.d.).
- **Confirmed payment infrastructure:** EPS supports transfers between banks and wallets. CBE reports that all eight participating institutions were connected by November 2025 (Central Bank of Eswatini [CBE], n.d., 2026, p. 70).
- **Proposed feature:** The sources reviewed do not confirm that an FNB eWallet can currently send directly to MoMo. Clearly label "Send to MoMo" as a *proposed simulation option*, rather than an existing FNB menu item.
- **Earlier system design:** CBE's 2021 specification separates immediate processing of payment instructions (clearing) from later settlement between institutions through SWIPSS. This helps explain the prototype; it does not confirm today's settlement schedule or bank arrangements (CBE, 2021).

FNB checks the sender's identity and permission to spend. EPS checks and routes the payment request. MTN checks whether the recipient wallet can receive it. CBE regulates the payment system and operates EPS; it does not manually approve each customer's transfer. A student application cannot debit a real eWallet merely because EPS or developer APIs exist.

1.2 Main development tasks

Area	Required features
Banking app	Login, dashboard, transfer form, review, approval, receipt and history.
USSD app	Two phone tabs with separate profiles, menus, PIN entry, SMS inboxes and account views.
Processing	Simulated FNB/EPS/MTN services; reserve funds, debit/credit once, record results and check interrupted payments.
Records	Account entries, transaction statuses, messages and checks that the services' records agree.

Useful terms: A *mock service* imitates a real service. A *quote* is the saved payment summary shown before approval. A *hold* reserves funds while a payment is being processed.

Where to look: System/data flow: pp. 2–3; screens/navigation: pp. 4–6; transfer implementation: pp. 7–10; phone updates/failures: pp. 11–12; checks/example: pp. 13–14; team tasks/references: p. 15.

2 System parts and responsibilities

Use one backend, a PostgreSQL database and two web interfaces. The backend is the server-side code that checks requests and manages records. Divide it into modules for the application, FNB, EPS and MTN. A background worker handles tasks such as message delivery and checking interrupted payments.

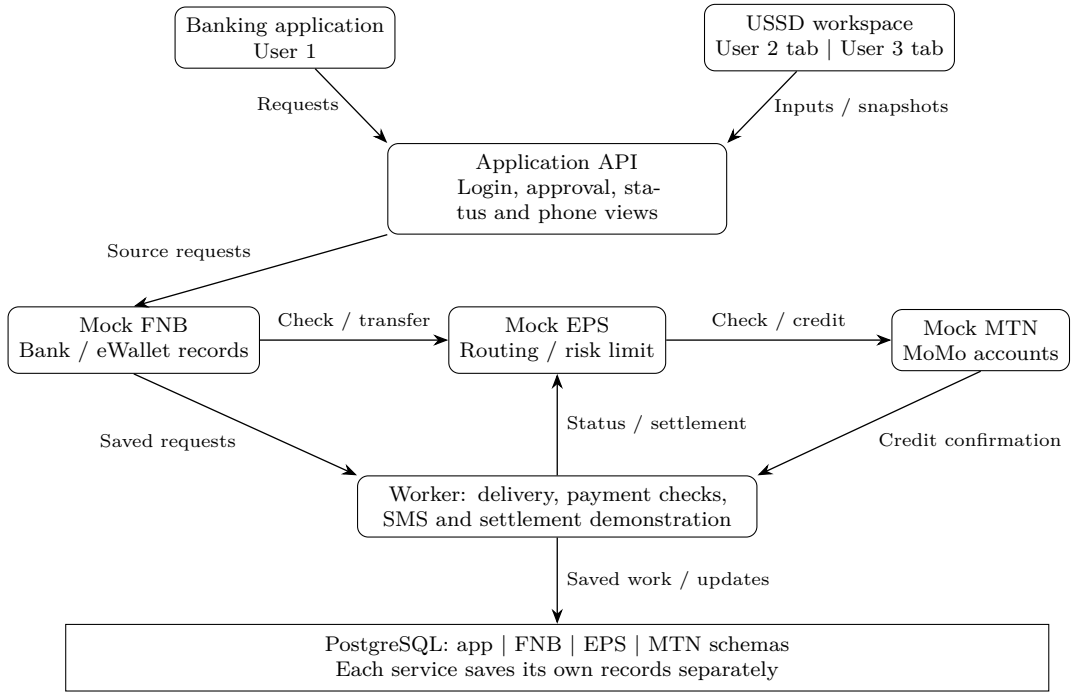


Figure 1. System architecture. FNB, EPS and MTN are simulated services. Replies and screen updates return through the application API.

Component	Main responsibilities
Application API	Manages login, phone sessions, user approval and account views. It must not directly change a provider's account balance. An *API* defines how software parts send requests to each other.
FNB service	Manages bank/eWallet accounts, wallet activation, sender checks, available funds, fees, limits, reserved amounts and final debits.
EPS service	Saves and routes requests, prevents duplicates, checks service identity and test risk limits, and records amounts owed between providers.
MTN service	Manages MoMo accounts, recipient checks, permitted name details, receiving limits, account credits and saved credit confirmations.
Background worker	Delivers saved messages, checks uncertain payments, compares records and runs the optional settlement demonstration.

2.1 Rules for connecting the parts

Use separate database schemas: named groups of tables for the application, FNB, EPS and MTN. Each module saves its own account entries and related messages in one *all-or-nothing database transaction*. If any part of that save fails, none of it takes effect. Do not treat separate providers as one shared transaction.

Normal payment requests follow: screen → application API → FNB → EPS → MTN. The browser must not contact the provider services directly. Allow test delays and failures between modules, even when they run in one program. Keep recovery, settlement and troubleshooting controls in a separate tester's panel, outside the phone menus.

3 How information moves through the system

This Level 0 Data Flow Diagram (DFD) shows the *whole prototype*, including its simulated financial services. It has no live connection to real FNB, MTN or EPS systems. Rectangles represent people; the rounded shape represents the prototype. Arrows show information being exchanged, not money travelling through the network.

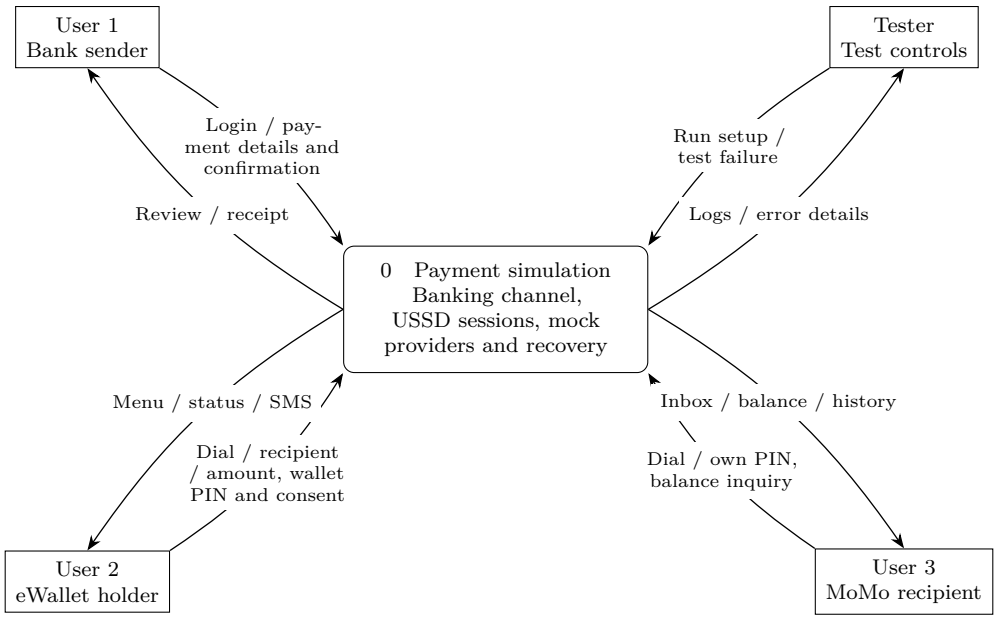


Figure 2. Level 0 DFD. Users enter their own details and receive only the views and messages they are allowed to access.

3.1 Information passed between services

From / to	Information sent	Expected result
Screen to FNB	Login/phone session, source account reference, recipient, amount, payment summary and approval.	Check permission and return a payment reference.
FNB through EPS	Test run/payment IDs, provider names, recipient reference, amount, currency, fee and time.	Save a decision linked to that exact request.
EPS to MTN	Payment ID, recipient reference, amount and currency.	Save either a credit confirmation or a final record that prevents a later credit.
Service to screen	Balance/status update, record version, SMS ID and receipt reference.	Refresh the relevant user's current view.

Keep credentials at the login or approval step. Do not place PINs, passwords or one-time codes in EPS/MTN payment messages, receipts or troubleshooting logs. The tester's panel must hide sensitive values.

A reply saying "request received" does not prove that money was credited. Show payment success only after the receiver credit and sender debit are confirmed. Record settlement between institutions separately. Page 9 lists the mock requests and responses that the programmer must implement.

4 UI/UX: banking screens and phone setup

Use these screen IDs in both wireframes and development tasks. Keep names, fields and navigation consistent. Each financial screen must show “Simulation” and the active test user’s identity.

ID / screen	Purpose, contents and input	Action / next screen
B01 Bank login	User 1 enters a test account ID and password. Show a clear error for invalid details. Do not display another customer’s account.	Correct details open B02. Failed login stays here; limit repeated attempts.
B02 Dashboard	Show the account name/reference, total and available balances, Send eWallet, History and Logout. Available balance is the amount the user can spend.	Send opens B03; History opens B07. Load balances from the backend.
B03 Send form	Select the source account; enter User 2’s test phone number and amount in Emalangeni. Put errors beside the affected field.	Continue requests a payment summary and opens B04; Back returns to B02.
B04 Payment review	Show recipient number, amount, fee and total debit. Display a recipient name only when verified.	Edit replaces the old summary. Confirm opens B05. Cancel sends nothing.
B05 Payment approval	Enter a one-use test code from User 1’s banking message tray. The code applies only to the displayed payment. Show expiry and retry guidance.	Submit once; disable repeated clicks while waiting for B06. Backend checks still prevent duplicates.
B06 Result	Show status, amount, recipient, time and payment reference. If the result is still unknown, provide Check status. Open the receipt or history.	Do not suggest resending an uncertain payment.
B07 History / detail	List recent payments first, with type, amount, recipient, status and reference. Open a payment to view its receipt.	Refresh from saved records; return to the dashboard.
D00 Phone setup	Provide two fixed slots: User 2/eWallet and User 3/MoMo. Select prepared test profiles; show each phone number/provider and test run ID. Do not display PINs. Open the USSD app.	Selecting a profile does not authorise a payment. Changing it ends that phone’s menu session.
F00 Tester panel	Choose a test scenario, start/reset a run, trigger service failures, inspect safe logs and run settlement. Keep these controls outside customer menus.	Reset starts a new test run. Do not allow direct balance editing.

4.1 General design rules

Keep valid inputs when a user corrects a form. Always display the entered number during review; an address-book name is not verified identity. If the summary expires or the amount/fee changes, require review and approval again. Logging out must not cancel an already submitted payment.

Use visible labels, clear keyboard focus and readable text. Explain outcomes using words and icons, rather than colour alone. The banking app can use a modern layout. Keep the phone’s USSD area as plain text and numbered choices.

5 UI/UX: phone menus and payment results

The app has two tabs: **User 2 – eWallet** and **User 3 – MoMo**. Each phone has a number header, menu history, reply field, Send/Reply, Back where allowed, End session and an SMS inbox. Switching tabs must preserve the other phone's screen and inputs.

ID / screen	Purpose, contents and input	Action / next screen
D01 Dial / activate	Enter the configured demo dial code. If User 2's eWallet is not activated, request the simulated SMS activation code and a new PIN. Otherwise request the phone user's own PIN before private account access.	Check with the simulated provider. Invalid code/PIN stays here and shows remaining attempts.
D02 U2 eWallet menu	Proposed menu: 1 Send to MoMo; 2 Balance; 3 Recent transfers; 0 Exit. Show the active number and provider.	Option 1 opens D03. Require a valid wallet login for balance/history.
D03–04 Transfer inputs	D03: enter the recipient number. D04: enter a positive amount with at most two decimal places. Show "Checking recipient" while verification runs.	Open D05 only after the recipient and payment summary pass checks. Keep invalid input at the relevant step.
D05 Recipient review	Show MoMo provider, full entered number, permitted partly hidden recipient name, amount, fee and total. Offer Confirm / Edit / Cancel.	Confirm opens D06. Edit creates a new summary. Block payment if required identity verification is unavailable.
D06 Wallet PIN	Enter User 2's five-digit demo PIN to approve this payment. Keep the approved summary visible. Hide the PIN and never add it to menu history.	A correct PIN submits one request to D07. An incorrect PIN reserves or transfers no funds.
D07 Processing / result	Show payment reference, available funds and amount reserved. Show a receipt for success, or explain a refusal. Offer Check status for PENDING/UNCERTAIN.	Home closes the menu only; it does not undo payment.
D08 U3 MoMo menu	After the recipient's own PIN: 1 Balance; 2 Recent transactions; 0 Exit. Show a new credit after MTN saves it.	Load balance/history from MTN. Receiving funds requires no recipient PIN or approval.
D09 SMS inbox	Show sender, text, time, payment reference and unread badge. Opening a message marks it read. Do not include PINs in messages.	Allow access even without an active USSD menu. Return to the previous menu position.
D10 Payment detail	Show amount, parties, status, time and receipt reference. U2 sees their own eWallet balance; U3 sees their own MoMo balance.	Refresh after re-entry. Do not use an SMS alone as proof of payment status.

5.1 Consistent messages

Success: "E150 sent to [number]. Ref [ID]." **Pending:** "Processing. E150 reserved." **Uncertain:** "Outcome being checked. Do not send again." **Rejected:** "Payment rejected. Reserved funds released." Show the last message only after the release is saved. Field errors must preserve safe inputs. If recipient checking times out, say verification is unavailable; do not claim the account is inactive.

6 Navigation and USSD session rules

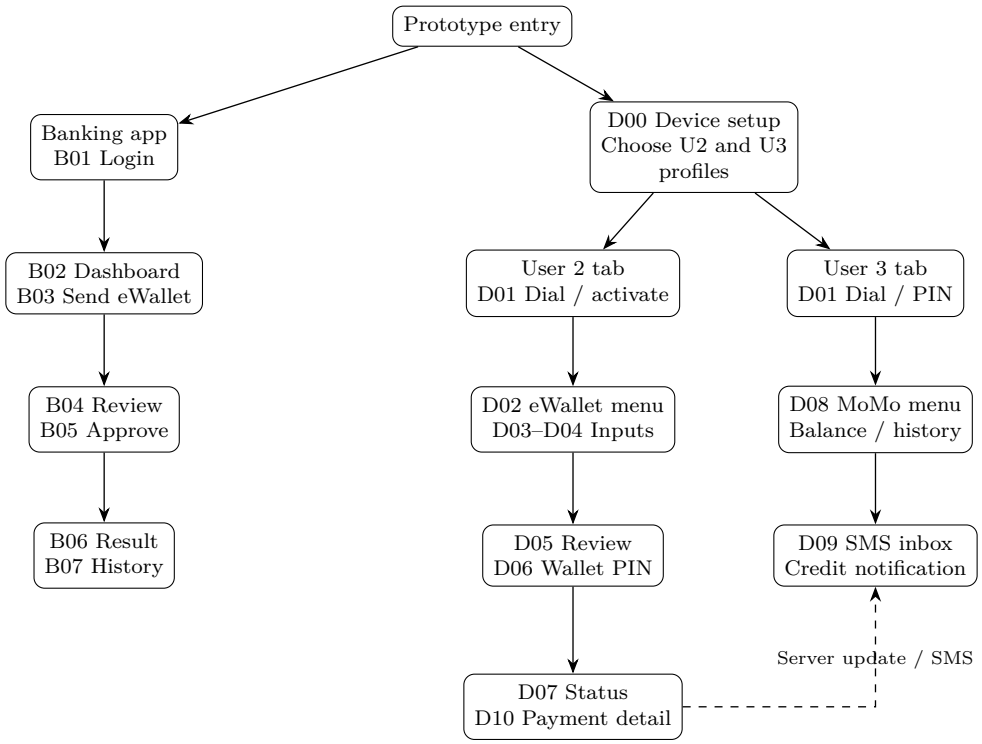


Figure 3. Screen navigation. The dashed arrow shows a notification from the backend; the phones do not transfer money directly to each other. Both SMS inboxes remain accessible.

6.1 How the menus should behave

- Use FNB's documented `*130*392#` as the eWallet dial example (FNB, n.d.). MTN's local published material uses `*007#` (MTN Eswatini, 2020). The following menu choices, PIN rules and "Send to MoMo" option are **chosen for this simulation**; they are not verified live menu behaviour.
- Keep a separate session for each phone and test run. Save the current screen, unfinished inputs, quote ID, login expiry and any submitted payment ID. Never save the entered PIN. An invalid reply should show the valid choices without moving to the next step.
- Close the menu after 120 seconds without activity. End/timeout clears unfinished menu inputs and login permission, but does not cancel a submitted payment. The user can log in again and check it in history.
- Back from review allows edits and replaces the old quote. After submission, Back/Home changes the screen only. Closing a menu cannot cancel a payment that may already have reached another service.

7 Banking transfer: User 1 to User 2

7.1 Starting conditions and expected result

Start: User 1 chooses Send eWallet. They must be logged in, own an active bank account, and enter a supported test recipient and a positive amount. **Result:** one bank-account debit and one eWallet credit. User 2 does not need a bank account.

1. Find User 1's permitted source account using their backend login session. Do not accept an account owner ID from the browser as proof of permission. Convert the entered number to the format used by the test phone directory.
2. Check the amount in cents, available funds, limits and account status. Find or create the recipient eWallet in the FNB service. Save a quote valid for 60 seconds with source, number, amount, fee and currency.
3. Show B04, then request the one-use approval code on B05. The code applies to that quote only. If details change or the quote expires, repeat review and approval.
4. In one FNB database transaction, check the duplicate-prevention key and lock the bank/eWallet records in a fixed order. This prevents competing requests from changing them at the same time. Recheck funds and limits, then save the balanced account entries and result.
5. In the same transaction, save outgoing messages for User 1's receipt and User 2's eWallet SMS. This saved message queue is called the *outbox*. If the save fails, undo all changes in that transaction.
6. Return SUCCESS with the FNB reference. If the reply is lost, check the original payment ID. A repeat using the same key must return the saved result, without another debit or credit.

7.2 Activating the eWallet

A newly created eWallet has recorded funds but cannot yet be used for spending. User 2 opens D01 on their phone tab, enters the activation code from that phone's simulated inbox, and chooses a demo PIN. FNB checks the run, phone, number, code expiry and whether the code was already used. Store a *salted PIN hash*: a protected value used to check the PIN without saving it as readable text.

Activation does not create a bank account. Selecting a phone profile must not skip activation or PIN checks. The simulated SMS code demonstrates the checking process; it does not verify a real SIM card. Treat activation rules chosen for this app as test rules, rather than confirmed production procedures.

7.3 Account entries and errors

Case	Required behaviour
E200, zero fee	Reduce the bank-deposit liability by E200 and increase the eWallet liability by E200. User 1's balance becomes E800; U2's becomes E200.
Sending fee f	Debit amount plus f from the bank account. Credit the transfer amount to the eWallet and record the fee separately. Demo default: $f = 0$.
Bad input / code	Show a clear field or approval error. Do not post account entries or create duplicate wallet records.
Insufficient funds / limit	Refuse before posting. Leave both balances unchanged.
Save or reply failure	Undo an incomplete save; otherwise recover the saved result. Failed notification delivery must not reverse payment.

This models a transfer within FNB, not FNB's actual internal design. Stage 1 creates no EPS/MTN credit or amount owed between institutions in this model.

8 Successful eWallet-to-MoMo transfer

Before starting: User 2 must control an activated eWallet. The test setting for the proposed MoMo option must be enabled. The recipient, amount and fee must pass the required checks. All services shown below are simulated.

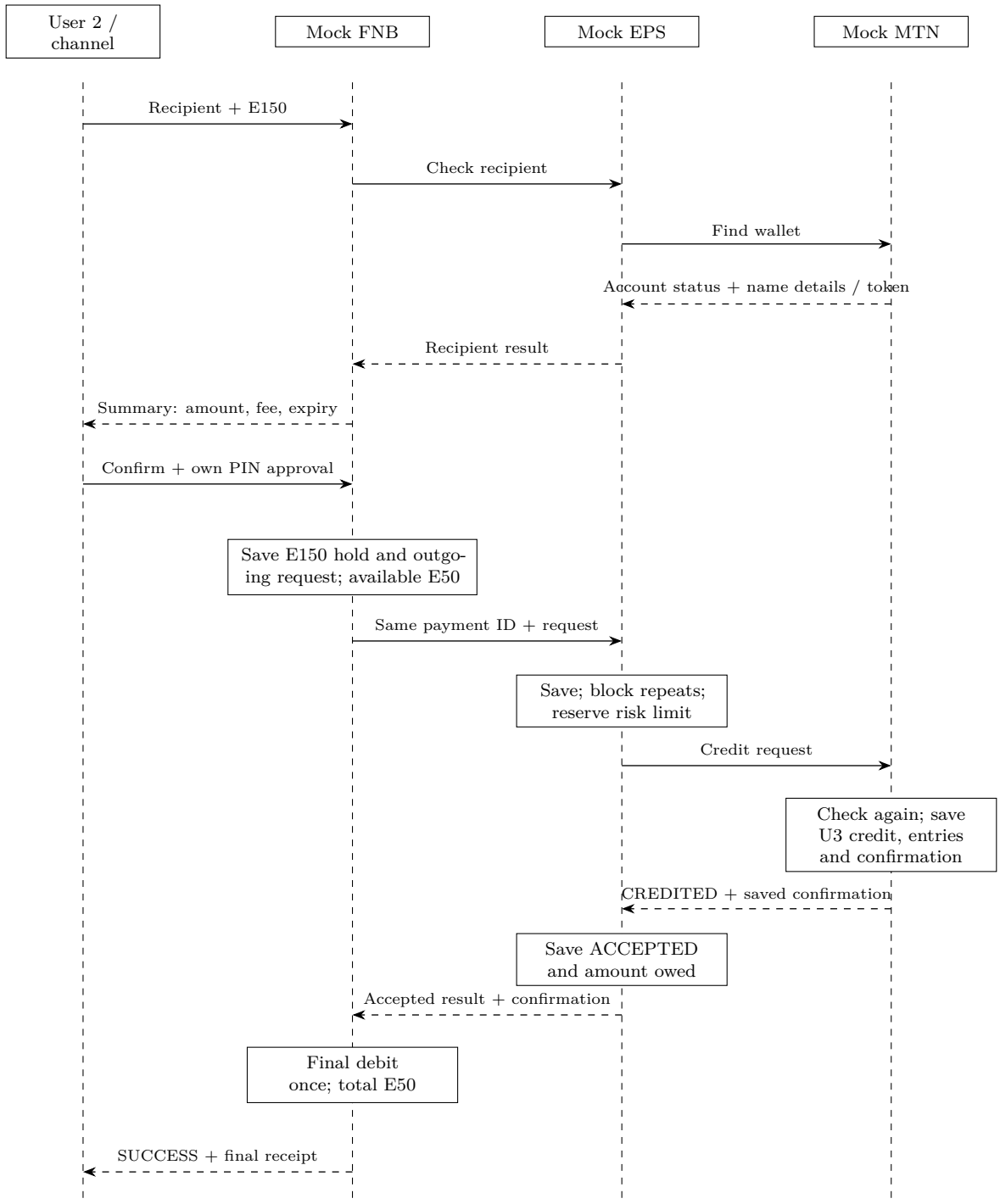


Figure 4. UML sequence diagram. Solid arrows show requests; dashed arrows show replies. Boxes on the vertical lines show work done inside a service. Read from top to bottom.

The User 2 column includes the application screen and its API request. Normal payment requests go through FNB, not directly to EPS. When MTN saves User 3's credit, it also queues the receiver update and SMS; page 11 explains how the phone tab receives them. Show SUCCESS only after confirming both the receiver credit and sender debit. Settlement between institutions is a separate step.

9 Requests and transfer processing

9.1 Required processing steps

1. **Prepare the payment:** check the number, find the MoMo account, obtain permitted partly hidden name details and a recipient reference, then save an expiring quote. An active account may still be unable to receive the chosen amount (MTN MoMo, n.d.). Check again before crediting.
2. **Reserve funds at FNB:** check the PIN against the approved quote. In one transaction, check the key, lock the account/daily-limit record, recheck ownership, product permission, funds and limits, reserve amount plus fee, and add the request to the outbox. Available funds decrease; the total recorded balance does not yet decrease.
3. **Send through EPS:** save the fixed payment details, verify the sending service, block duplicates and reserve space within the test risk limit. Forward the same payment ID to MTN. This institutional limit is separate from the customer's reserved funds.
4. **Credit at MTN:** in one transaction, recheck the account and receiving limit, check the unique incoming payment key, and save one credit, its confirmation and outgoing messages. Return the saved confirmation for a repeat request.
5. **Complete the payment:** after confirmed credit, EPS saves ACCEPTED and the amount owed between providers. FNB makes the reserved amount a final debit, updates daily usage and saves SUCCESS. After a failure, complete any missing step once instead of repeating the whole payment.

9.2 Mock API requests

These paths, fields, response codes and timing rules are chosen for the prototype. They are not official FNB, EPS or MTN APIs. Use integer cents and currency SZL.

Request	Input	Response / rule
`POST /sim/bank/quotes`	Source, test number, amount	Quote and fee; no balance change.
`POST /sim/bank/payments`	Quote, request key, approval proof	Saved Stage 1 result or PENDING.
`POST /sim/ussd/reply`	Phone session ID, reply	Next menu; consume PIN input at authentication.
`POST /sim/wallet/quotes`	eWallet, recipient, amount	Recipient reference, name details, total and expiry.
`POST /sim/wallet/payments`	Quote, request key, PIN proof	Payment ID. "Request received" is not SUCCESS.
`GET /sim/payments/ID`	User/phone permission	Current status, receipt and reserved amount.
`GET /sim/devices/ID/view`	Phone permission	Own menu, permitted account view and inbox position.
`GET /sim/events`	Run/phone scope, last event ID	Update notice; regular polling may be used instead.

Errors: return a fixed reason code, safe message, affected field if any, whether retry is allowed, and any existing payment ID. Distinguish invalid input/PIN, ineligible recipient, funds/limit refusal, unavailable service and unknown outcome. A network error does not prove that no credit occurred.

Internal requests: EPS needs lookup, transfer and status operations. MTN needs lookup, credit, status and final no-credit confirmation. Only identified backend services may call these; the UI must not use them to skip FNB approval.

Message fields: run/payment IDs, request key, source/destination provider, source account reference, recipient token, cents/fee, currency, quote and time. Replies add switch/credit references, status, reason and version. Exclude PINs/OTPs. Keep a payload hash to detect changed details. The *idempotency key* prevents repeats: the same key/details returns the saved result; changed details with that key are refused.

10 Database records and balance updates

Group tables by the service responsible for them, even within one PostgreSQL database. Every record must belong to a test run. Include the run ID in related-record checks so that delayed messages cannot update another run.

Record	Main fields / links	Required rule
Run / phone	Run ID, scenario, U2/U3 phone slot, provider and test profile	Keep the phone role fixed; reset creates a new run.
User / login	User ID, role, password/PIN hash, failed-attempt count	Own-account access only; no readable saved secrets.
Account / wallet	ID, provider, product, owner, number, currency, status and limit tier	Keep bank and eWallet accounts separate. U2 needs no bank account.
Menu session / quote	Phone, screen, safe draft, login expiry; quote details, recipient token and expiry	No PIN in drafts; approved details cannot change.
Transfer / status history	Payment ID, stage, parties, key, hash, status/version; dated status changes	Unique request key; keep earlier statuses.
Hold / daily limit	Account, reserved amount/fee, hold status; reserved and spent daily totals	Lock/check together; complete or release a hold once.
Journal / entries	Service, payment, entry type, account, debit/credit cents	Balance debit and credit totals; prevent repeated entries.
EPS request / amount owed	Providers, fixed details, decision, risk reservation, credit reference and settlement batch	One amount-owed record per accepted payment.
MTN credit / no-credit record	Incoming key, recipient, amount, result and time	A payment cannot have both a credit and a final record preventing credit.
Outbox / inbox / SMS	Event, payment, phone, run, delivery state, attempts; message text/read flag	Save messages reliably and process repeats once.
Record-check case	Payment IDs, mismatch, evidence, assigned person and resolution	Add linked corrections; never overwrite account history.

10.1 Total, reserved and available funds

The *booked balance* is the total recorded balance. A *hold* is money reserved for a payment. The *available balance* is what remains spendable:

$$\text{Available balance} = \text{Booked balance} - \text{Active holds}.$$

For E200 total and E150 reserved, E50 is available. To finish the debit (*capture the hold*), reduce the total to E50 and remove the hold in the same save. Do not debit twice. A final rejection releases the hold and leaves E200 total. Never release an uncertain payment's hold merely because a timer expires.

Balanced entries: a journal records matching debits and credits. Stage 1: debit bank-deposit liability, credit eWallet liability. MTN credit: debit the amount due from its clearing agent, credit MoMo liability. FNB's final debit: debit eWallet liability, credit the amount owed for clearing. A *liability* is money the provider owes; a *receivable* is money owed to the provider. Each service saves its own balanced entries.

Settlement: later, the institutions settle amounts owed using fictional settlement accounts and a receiving agent. *Netting* offsets payments in opposite directions. It does not credit the customer again. Keep UNSETTLED/BATCHED/SETTLED/EXCEPTION separate from customer SUCCESS; real production arrangements still require confirmation.

11 Keeping both phones updated and sending SMS

11.1 Use the backend as the source of account information

The tabs represent two phones, but they must not keep separate copies of the financial database. User 2 submits a request; the simulated services process it; the backend then updates User 3's view. Do not increase MoMo balances in browser code or send payment value directly between tabs.

1. Start a test run and assign a fictional profile to each phone slot. Keep separate phone session IDs and permissions. Profile selection shows the test identity; it does not give permission to spend or view protected balances.
2. Keep separate menus and unfinished inputs for U2 and U3. Switching tabs changes only which screen is visible. It must not reset the hidden tab, send a reply again or cancel work.
3. Save account changes and outbox messages together. An MTN credit event contains `event_id`, `run_id`, `device_id=U3`, `payment_id`, `type=CREDIT_POSTED`, account version and time. An *event* is a saved notice that something changed; it contains no credentials.
4. Listen for both phones' updates using server-sent events (SSE), including the hidden tab. Alternatively, poll both views every two seconds. SSE sends notices from server to browser; polling asks repeatedly for updates. On a notice, load the current permitted view instead of applying an old event's balance.
5. Ignore repeated event IDs and older account versions. Save the last event position. After reconnecting, replay missed events or load a fresh view before continuing.
6. Update the hidden phone's unread badge while connected. Refresh its inbox/account view when opened. User 3 may receive the credit SMS while FNB is still completing the sender's debit; show each side's actual status.

11.2 Simulated SMS messages

Recipient	Send after	Example message
User 1	Stage 1 entries are saved	"E200 sent to [U2 number] via eWallet. Ref S1-001."
User 2	Stage 1 eWallet credit is saved	"You received E200 in eWallet. Ref S1-001." If activation is needed, direct the user to D01.
User 2	Stage 2 final debit is saved	"E150 sent to [U3 number]. Fee E0. Available E50. Ref S2-001."
User 3	MTN credit is saved	"MoMo: received E150. Balance E150. Ref S2-001."
User 2	Rejection and hold release are saved	"Payment S2-001 rejected. E150 reservation released."

The worker creates one message for each recipient/message type from saved events. Prevent duplicates using (run, payment, recipient, type). Reading changes only the read flag. Retry failed SMS delivery separately; do not repeat or reverse the payment.

11.3 Refresh, disconnection and reset

Save financial records and inboxes on the server. Refresh should restore the phone session or require its own PIN again; it must not submit a new transfer. When offline, show "Offline – balances may be outdated", block new payment submissions and keep the last reference for later checking.

A tester reset creates a new run ID and clears displayed sessions. Old messages, service replies and workers must stay linked to the old run. They cannot credit the new one. The two tabs imitate phones for teaching; they do not provide real device or telecom isolation.

12 Payment statuses and failure handling

Status	Meaning and screen behaviour
PENDING	Funds are reserved and processing continues. Show amount/reference; do not offer Send again.
UNCERTAIN	The request may have reached another service, but its result is unknown. Keep funds reserved; offer Check status.
SUCCESS	Receiver credit, EPS acceptance and sender debit are confirmed. Show the receipt.
REJECTED / FAILED NOT SENT	Credit was safely ruled out, or an unsent request was cancelled. Announce released funds only after the release is saved.

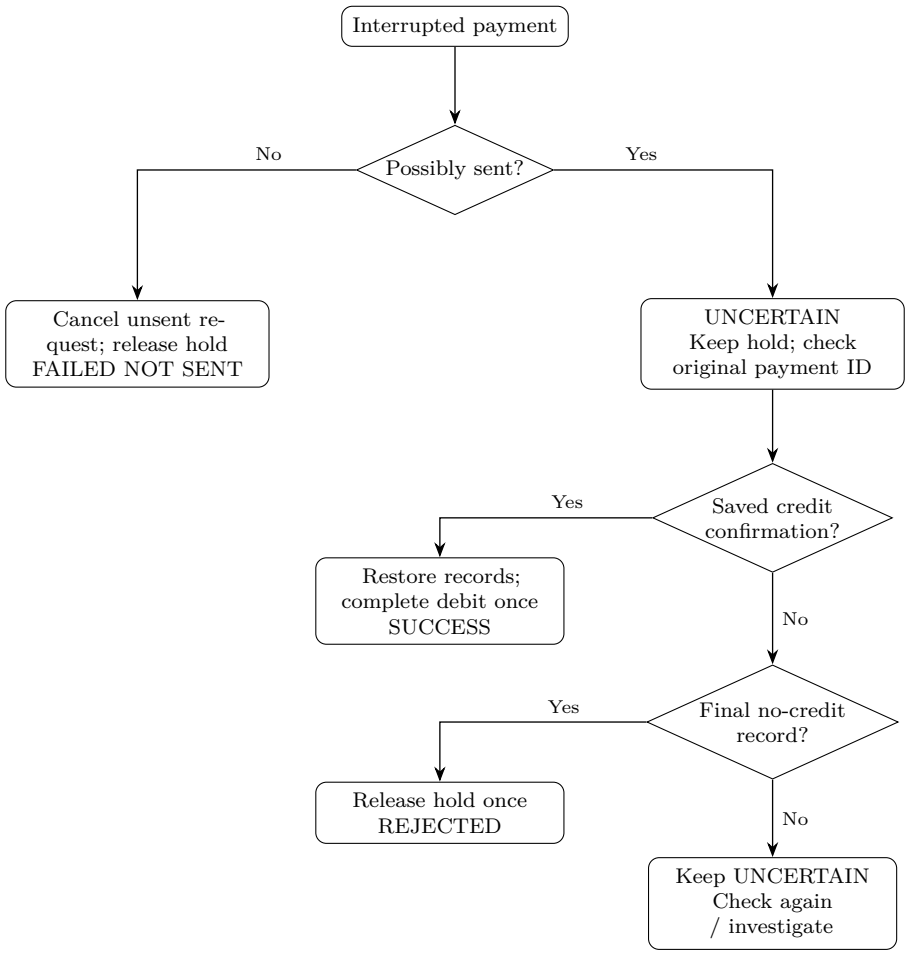


Figure 5. Checking an interrupted payment. A timeout alone does not justify returning reserved funds. A confirmed refusal before credit can go directly to the final no-credit step.

Prevent conflicting actions: cancel an outbox item only while it is READY, before a sending worker takes it. Once it may have been sent, check the original payment ID. A final MTN no-credit record must also block any delayed request from crediting that ID. NOT FOUND alone is insufficient. If a saved credit exists, recover the confirmation and complete FNB's debit once; never release that payment's hold.

Recovery rules: retry communication with the same ID and longer gaps between checks. Refer prolonged uncertainty for investigation, without automatic refund. A completed payment to the wrong person requires support; an approved return must be a separate linked transaction. Settlement failure must not undo the customer's spendable credit.

13 Security, test settings and checks

13.1 Checks enforced by the backend

Check account ownership for every change and private view. A phone/profile ID from the browser is not permission to debit. Use salted password/PIN hashes, limit failed attempts across sessions and never print PINs in menu history. Approval must match source, recipient, amount, fee, currency, quote expiry and run. Use it once and allow only one payment per quote, even with a new request key.

Use encrypted network connections (TLS), identified internal services and verified service replies. Hide sensitive details in logs. Match receipt ID, amount, currency, recipient and version before updating a payment. Fake, late or repeated messages must not create extra entries. Use fictional data and keep secrets out of browser storage and source code.

13.2 Settings chosen for the demonstration

Setting	Prototype default
Currency / fees	SZL; amounts in integer cents; both fees E0.
Wallet limits	E1–E10,000 per transfer; E50,000 per day; maximum receiving balance E50,000. Apply all relevant limits.
EPS risk limit	E500 in the example. Count reserved requests plus accepted payments awaiting settlement.
PIN / code / menu	Five-digit demo wallet PIN. One-use code and quote expire after 60 seconds; an inactive menu closes after 120 seconds.
Attempts / waiting	Three failed attempts cause a 60-second pause. After five seconds waiting for payment, show status. Refer unresolved cases after five minutes.

These are test settings, not confirmed FNB/MTN/EPS rules, fees or promised response times. A timeout starts checking, not release of funds for a possibly sent payment. Each service must check and save account capacity, available funds, daily reservations or institutional risk usage together in its own database transaction.

13.3 Required acceptance tests

ID	Test	Expected evidence
T01	Send E200 to eWallet, then E150 to MoMo	E800/E50/E150; balanced entries and matching receipts.
T02	Repeat submit; change key for same quote; change details with same key	One debit, credit and amount-owed record. Refuse conflicting requests.
T03	Two E150 requests from E200 at the same time	At most one succeeds; available balance stays non-negative.
T04	Inactive recipient; recipient blocked after checking	No credit; release reserved funds only after final rejection.
T05	MTN saves credit but reply is lost; crash before final debit	UNCERTAIN becomes SUCCESS; no refund or duplicate credit.
T06	Wrong PIN; another user's account/phone; fake service reply	No unauthorised entries; clear error and audit record.
T07	Switch tabs, refresh, reconnect; repeat an SMS event	Screens recover current information; one SMS and one credit.
T08	Reset before an old reply arrives; retry settlement	Old work cannot change the new run; settle once.

14 Worked example: E200 followed by E150

Prepare one test run: User 1 owns BANK-U1 with E1,000. User 2's EW-U2 has E0 and needs activation. User 3's MM-U3 is active, has E0 and a prepared demo PIN. Both fees are E0. Phone profiles use fictional directory entries, never real network destinations. Below, U2 and U3 stand for their configured test numbers.

1. **Banking form:** User 1 logs in, selects Send eWallet and enters U2's number and E200. FNB saves quote Q1: amount E200, fee E0, total E200.
2. **Review and approval:** User 1 approves Q1 with the one-use test code. FNB saves S1-001 once. Bank balance becomes E800 and eWallet balance E200. B06 shows success; U2 receives the eWallet SMS.
3. **Wallet activation:** On the User 2 tab, enter the simulated activation code and choose a PIN. User 2 can now spend from EW-U2. No bank account is created.
4. **USSD inputs:** Open the eWallet menu, choose the proposed Send to MoMo option, and enter U3's number and E150. Check L2 finds an eligible MM-U3 and returns a permitted partly hidden name and recipient token DT3.
5. **Recipient review:** D05 shows U3's number/provider, name details, E150, E0 fee and E150 total. User 2 confirms and enters their own PIN on D06. User 3 does not enter a PIN to receive the transfer.
6. **Reserve and send:** FNB saves hold H2 and outgoing request S2-001 together. EW-U2 has E200 total but E50 available. EPS saves S2-001 and reserves E150 within its E500 test risk limit.
7. **Credit the recipient:** MTN checks MM-U3 again, saves the E150 credit and confirmation M3-R1, and queues U3's update/SMS. The receiver tab shows an unread badge and refreshes the permitted balance view to E150.
8. **Complete the sender debit:** EPS saves ACCEPTED and amount-owed record O2. FNB completes H2 once, leaving E50 total/available, then sends SUCCESS to D07. U2 receives the final receipt/SMS. Both histories keep S2-001.
9. **Demonstrate settlement:** settle O2 in batch B1 using fictional institutional accounts. FNB settlement balance changes E1,000 → E850; receiving agent changes E500 → E650. Customer balances stay E800/E50/E150.

14.1 Expected balances

Point	Bank U1	EW total	EW available	MoMo U3
Initial	E1,000	E0	E0	E0
After Stage 1 / activation	E800	E200	E200	E0
Stage 2 funds reserved	E800	E200	E50	E0
MTN credited; source not finished	E800	E200	E50	E150
Sender debit / later settlement	E800	E50	E50	E150

Before the sender debit finishes, do not add total recorded balances as though they were all spendable. The E150 hold still blocks reuse. If the recipient was credited, recovery must complete that debit rather than release the hold.

14.2 Repeat with controlled failures

Inactive U3: checking fails; no funds reserved; U2 keeps E200. **Late refusal:** confirmed no-credit and hold release leave U2 with E200. **Switch outage after possible sending:** show UNCERTAIN and E50 available. Finish at E50/E150 if credited, or release funds only after a final record blocks later credit. Do not repeat or reverse Stage 1; User 1 stays at E800.

15 Team checklist and references

15.1 Suggested development order

Agree screen IDs, statuses, test accounts and payment fields first. Build Stage 1 and activation, then Stage 2 success. Next test duplicate requests, competing payments, interrupted transfers, phone updates and SMS delivery. Give the tester access to account-entry traces, but keep technical controls out of customer menus.

Area	Output for the four-member team
Programming	Provider services, request formats, database rules, background worker and tests T01–T08.
UI/UX design	Wireframes B01–B07 and D00–D10; separate F00 tester panel; review all result messages and navigation.
Integration / testing	Prepared runs showing both phones, failures, repeated/simultaneous requests, reconnection and reset.
Documentation / coordination	Keep screen/message IDs consistent; list assumptions and questions needing institutional confirmation.

Before any real integration: ask FNB whether eWallet funds may be used for this transfer; ask MTN about receiving rules and permitted recipient details; ask the EPS operator about access, message formats, when payments become final, returns, fees, limits and settlement. These matters remain unconfirmed for this exact product.

If the direct route is unsupported, investigate an approved transfer from eWallet to the holder's own bank account followed by bank-to-MoMo, or bank-to-MoMo at the first sending stage. Confirm local availability. Requiring a bank account changes who can use the service and narrows the original purpose.

Editing: this LaTeX file includes all five diagrams (drawn with TikZ) and compiles independently with pdfLaTeX. This revision supplies the guide, not the HTML application.

15.2 References

Central Bank of Eswatini. (2021). *Implementation of the national payment switch of the Kingdom of Eswatini: Request for proposal*. Historical specification, Section IV, pp. 3–4, 20–21. https://www.centralbank.org.sz/wp-content/uploads/2021/04/Eswatini-National-Payment-Switch-RFP-v1.5.pdf

Central Bank of Eswatini. (2026). *Annual integrated report 2025/26* (p. 70). https://www.centralbank.org.sz/wp-content/uploads/2026/08/Annual-Integrated-Report-2025-2026-online-compress.pdf

Central Bank of Eswatini. (n.d.). *FAQs: Eswatini payment switch*. Retrieved October 15, 2026, from https://www.centralbank.org.sz/faq/

First National Bank of Eswatini. (n.d.). *eWallet*. Retrieved October 15, 2026, from https://www.fnbswaziland.co.sz/services/eWallet.html

MTN Eswatini. (2020, May 21). *MoMo bank transfers*. Historical local channel information; not evidence of the proposed eWallet menu. https://www.mtn.co.sz/momo-bank-withdrawals/

MTN MoMo. (n.d.). *Validate account holder* [Developer documentation]. Retrieved October 15, 2026, from https://momodeveloper.mtn.com/content/html_widgets/v98wn.html